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## Computer Networks Lab

### Week 04

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## Configuring DNS and Email Servers in Packet Tracer

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### Tools:

Cisco Packet Tracer 8.2+: Download from <https://www.netacad.com/resources/lab-downloads?courseLang=en-US>

### Overview:

This lab focuses on configuring a Domain Name System (DNS) server and an email server in Cisco Packet Tracer to enable name resolution and email communication in a simulated network. By setting up a network topology with PCs, servers, a switch, and a router, you will configure IP addresses, DNS services, email services (SMTP/POP3), and test email communication between clients. This lab complements the study of secure data transmission (e.g., HTTP vs. HTTPS) by demonstrating how DNS resolves email addresses and how email servers handle communication.

### Objectives:

1. Set up a DNS server to resolve domain names to IP addresses for email services.
2. Configure an email server with user accounts and enable SMTP/POP3 services.
3. Set up email clients on PCs to send and receive emails.
4. Test DNS resolution and email functionality to verify the configuration.

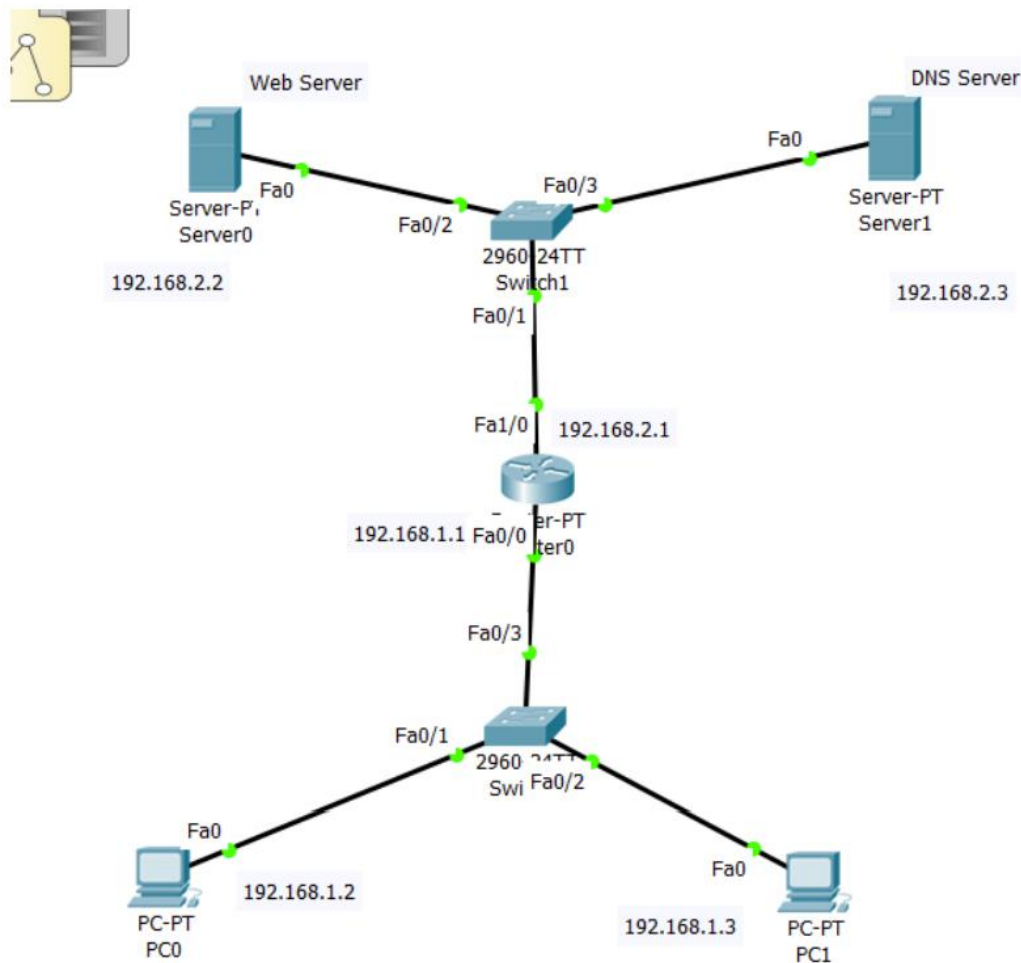
### DNS server configuration in Packet Tracer:

A Domain Name Service (DNS) server resolves host names into IP addresses. Although we can access a network host using its IP address, DNS makes it easier by allowing us to use domain names which are easier to remember. For example, it's much easier to access the Google website by typing <http://www.google.com> as compared to typing <http://172.217.21.14>. In either case, you'll access the Google website, but using a domain name is obviously easier. Now, before any host can use a DNS service, we must configure a DNS server first. For example, when you type the URL <http://www.google.com> in your browser, the host will query the DNS server for the IP address of <http://www.google.com>. The DNS server will resolve <http://www.google.com> into an IP address and then answer back the host with the IP address.

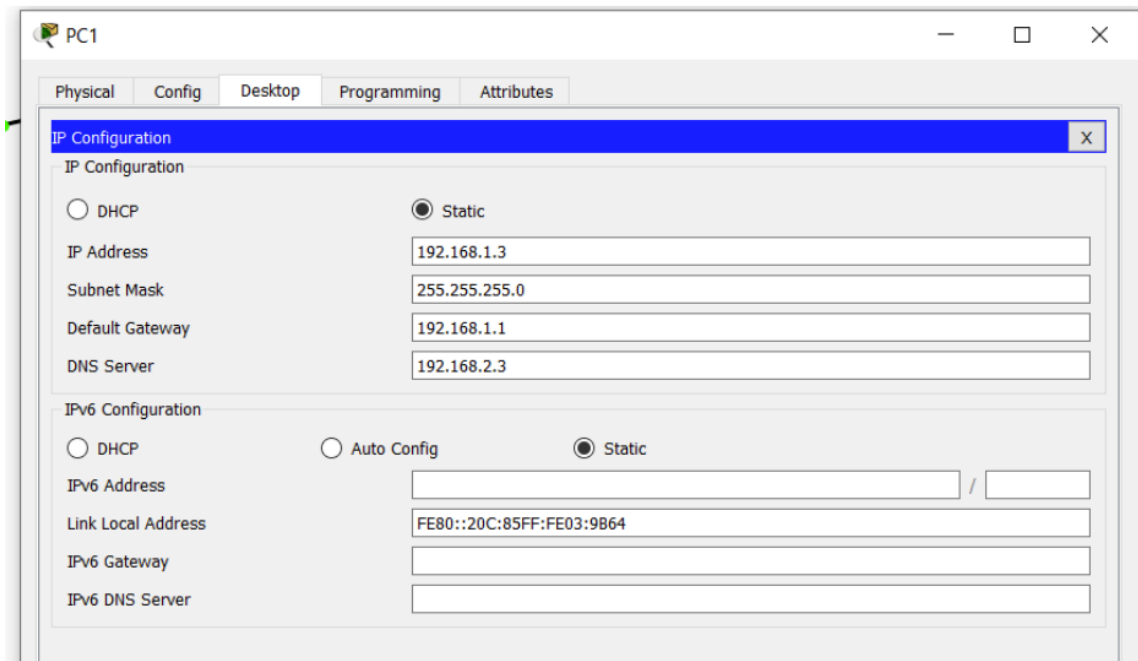
An HTTP server is a web server. It stores web resources that can be accessed by a web client. Your PC's browser(a web client) requests for web resources from a web server over the internet. Web resources are files such as text and images that the server will give to the client on request.

## Step by Step Configuration:

### 1. Network Topology:



### 2. Configuration of PC



PC1

Physical Config Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 192.168.2.3

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

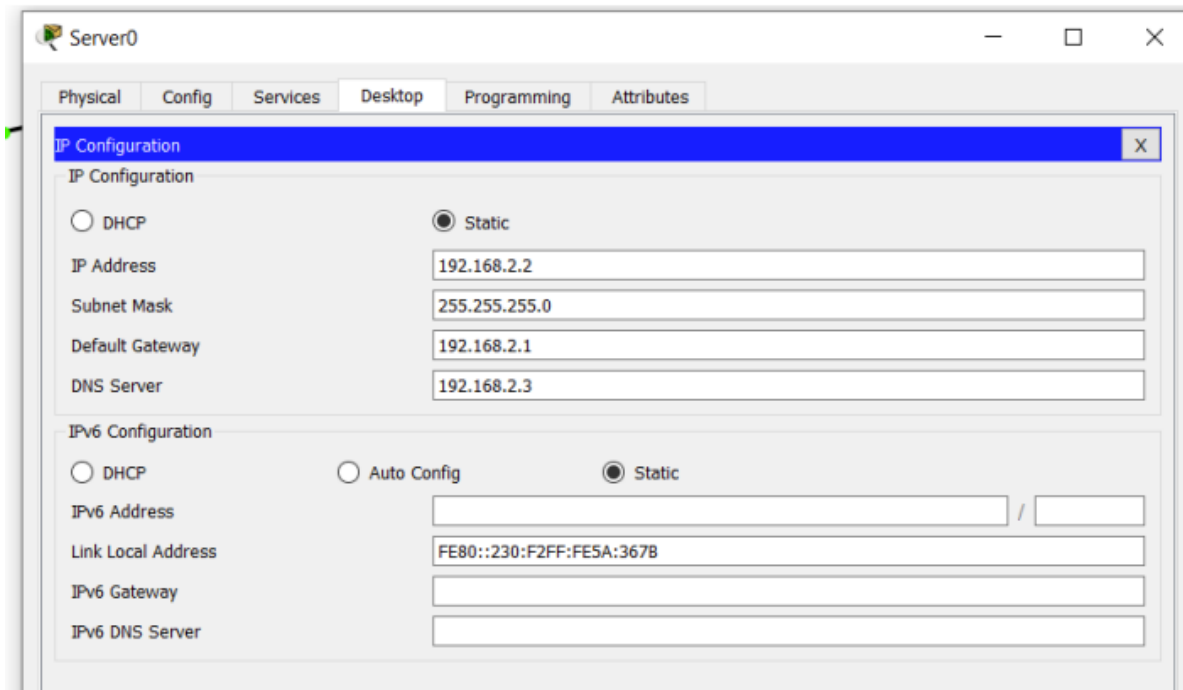
IPv6 Address: /

Link Local Address: FE80::20C:85FF:FE03:9B64

IPv6 Gateway:

IPv6 DNS Server:

### 3. Configuration of Web Server



Server0

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

DNS Server: 192.168.2.3

IPv6 Configuration

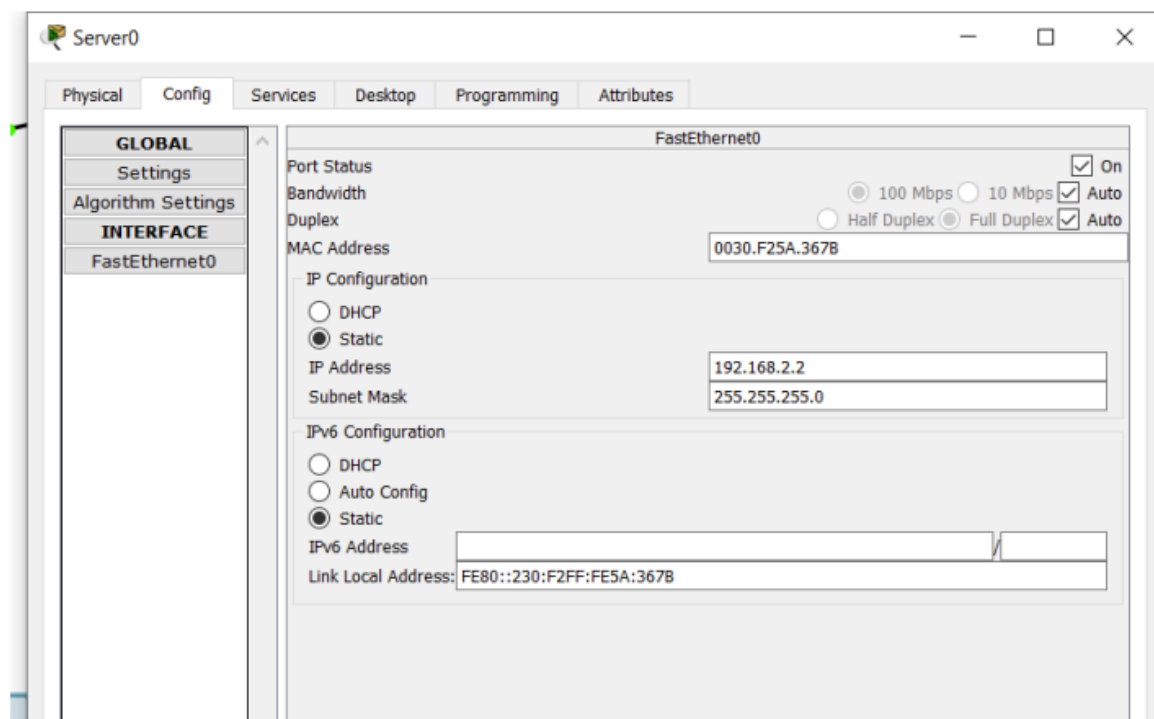
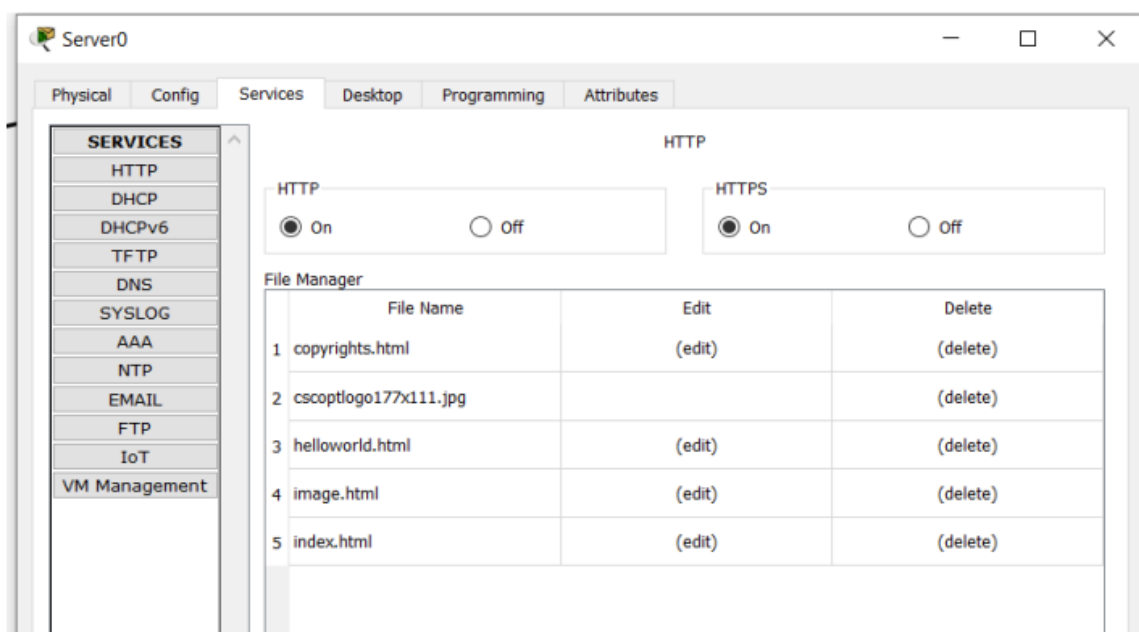
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

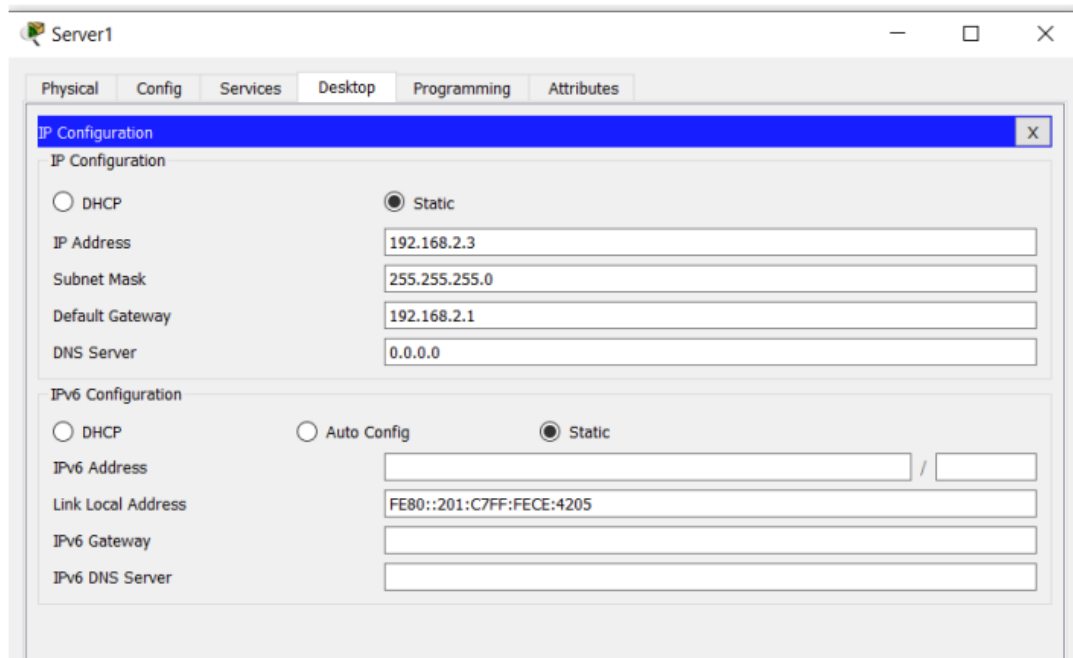
Link Local Address: FE80::230:F2FF:FE5A:367B

IPv6 Gateway:

IPv6 DNS Server:



#### 4. Configuration of DNS Server



Server1

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address: 192.168.2.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

DNS Server: 0.0.0.0

IPv6 Configuration

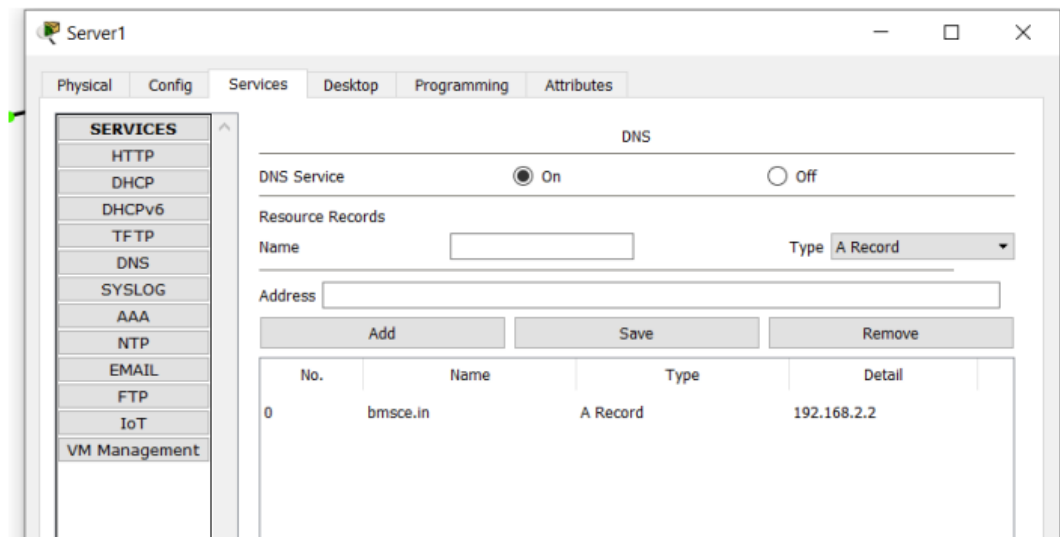
☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:C7FF:FECE:4205

IPv6 Gateway:

IPv6 DNS Server:



Server1

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management

DNS

DNS Service: ☒ On ☐ Off

Resource Records

Name: Type: A Record

Address:

Add Save Remove

| No. | Name     | Type     | Detail      |
|-----|----------|----------|-------------|
| 0   | bmsce.in | A Record | 192.168.2.2 |

## 5. Demo

1. Use Simulation Mode Select the PC with IP Address 192.168.1.3 . Click on the webBrowser and type 192.168.2.2 . Show the Demo of TCP by right clicking on the Envelop. Note down the values at every Layer.

PDU Information at Device: PC1

OSI Model    Outbound PDU Details

At Device: PC1  
Source: PC1  
Destination: 192.168.2.2

| In Layers | Out Layers                                                    |
|-----------|---------------------------------------------------------------|
| Layer7    | Layer 7:                                                      |
| Layer6    | Layer6                                                        |
| Layer5    | Layer5                                                        |
| Layer4    | Layer 4: TCP Src Port: 1027, Dst Port: 80                     |
| Layer3    | Layer 3: IP Header Src. IP: 192.168.1.3, Dst. IP: 192.168.2.2 |
| Layer2    | Layer 2: Ethernet II Header 000C.8503.9B64 >> 00D0.FFB6.234D  |
| Layer1    | Layer 1: Port(s): FastEthernet0                               |

1. The HTTP client makes a connection to the server.

## PDU information at Web server

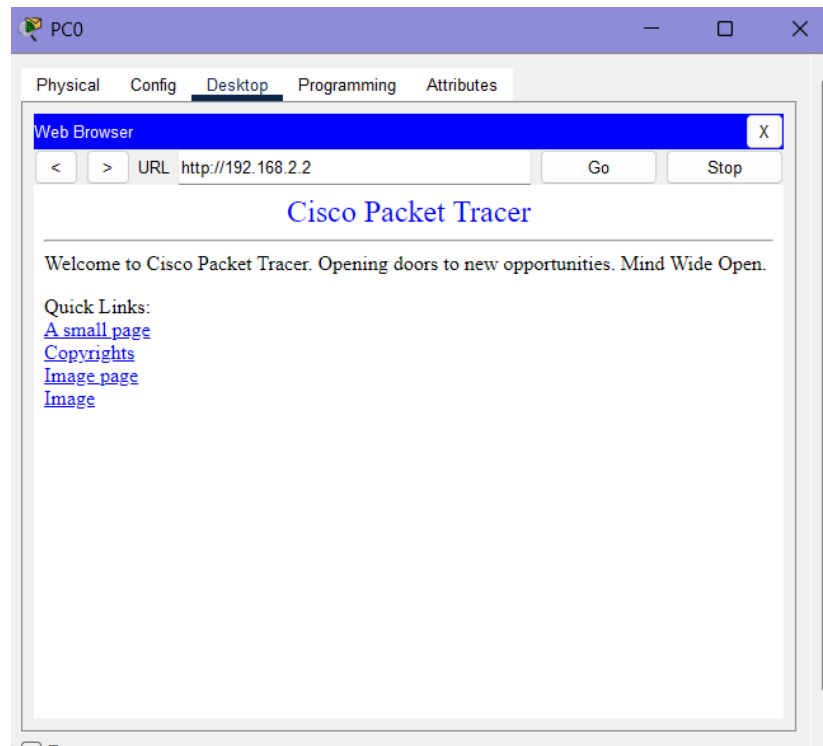
PDU Information at Device: Server0

OSI Model    Inbound PDU Details    Outbound PDU Details

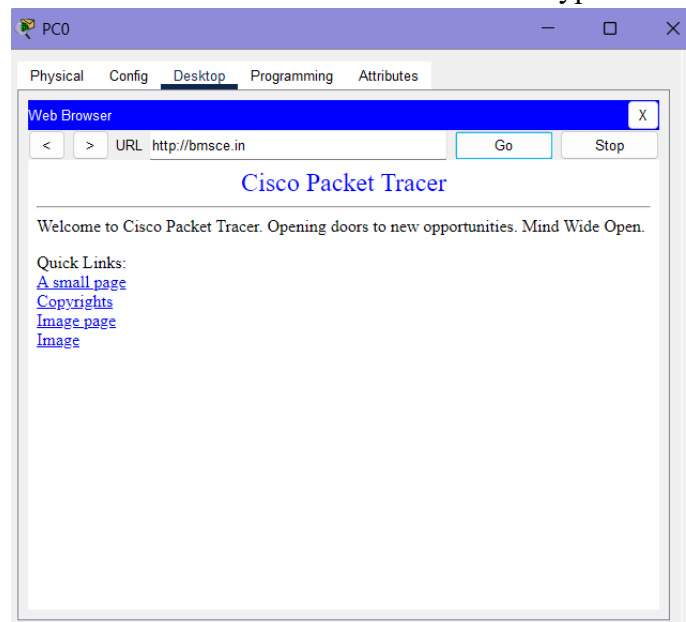
At Device: Server0  
Source: PC1  
Destination: 192.168.2.2

| In Layers                                                     | Out Layers                                                    |
|---------------------------------------------------------------|---------------------------------------------------------------|
| Layer7                                                        | Layer7                                                        |
| Layer6                                                        | Layer6                                                        |
| Layer5                                                        | Layer5                                                        |
| Layer 4: TCP Src Port: 1027, Dst Port: 80                     | Layer 4: TCP Src Port: 80, Dst Port: 1027                     |
| Layer 3: IP Header Src. IP: 192.168.1.3, Dst. IP: 192.168.2.2 | Layer 3: IP Header Src. IP: 192.168.2.2, Dst. IP: 192.168.1.3 |
| Layer 2: Ethernet II Header 00E0.F97B.607B >> 0030.F25A.367B  | Layer 2: Ethernet II Header 0030.F25A.367B >> 00E0.F97B.607B  |
| Layer 1: Port FastEthernet0                                   | Layer 1: Port(s): FastEthernet0                               |

1. FastEthernet0 receives the frame.



2. At the same PC in Simulation mode select web Browser and type bmsce.in



## PDU Information at Device: PC1

OSI Model

Outbound PDU Details

At Device: PC1  
Source: PC1  
Destination: 192.168.2.3

## In Layers

Layer7

Layer6

Layer5

Layer4

Layer3

Layer2

Layer1

## Out Layers

Layer 7: DNS

Layer6

Layer5

Layer 4: UDP Src Port: 1026, Dst Port: 53

Layer 3: IP Header Src. IP: 192.168.1.3,

Dest. IP: 192.168.2.3

Layer 2: Ethernet II Header 000C.

8503.9B64 &gt;&gt; 00D0.FFB6.234D

Layer 1: Port(s): FastEthernet0

## 3. At the DNS Server

## PDU Information at Device: Server1

OSI Model

Inbound PDU Details

Outbound PDU Details

At Device: Server1  
Source: PC1  
Destination: 192.168.2.3

## In Layers

Layer 7: DNS

Layer6

Layer5

Layer 4: UDP Src Port: 1026, Dst Port: 53

Layer 3: IP Header Src. IP: 192.168.1.3,  
Dest. IP: 192.168.2.3Layer 2: Ethernet II Header 00E0.F97B.  
607B >> 0001.C7CE.4205

Layer 1: Port FastEthernet0

## Out Layers

Layer 7: DNS

Layer6

Layer5

Layer 4: UDP Src Port: 53, Dst Port: 1026

Layer 3: IP Header Src. IP: 192.168.2.3,  
Dest. IP: 192.168.1.3Layer 2: Ethernet II Header 0001.C7CE.  
4205 >> 00E0.F97B.607B

Layer 1: Port(s): FastEthernet0

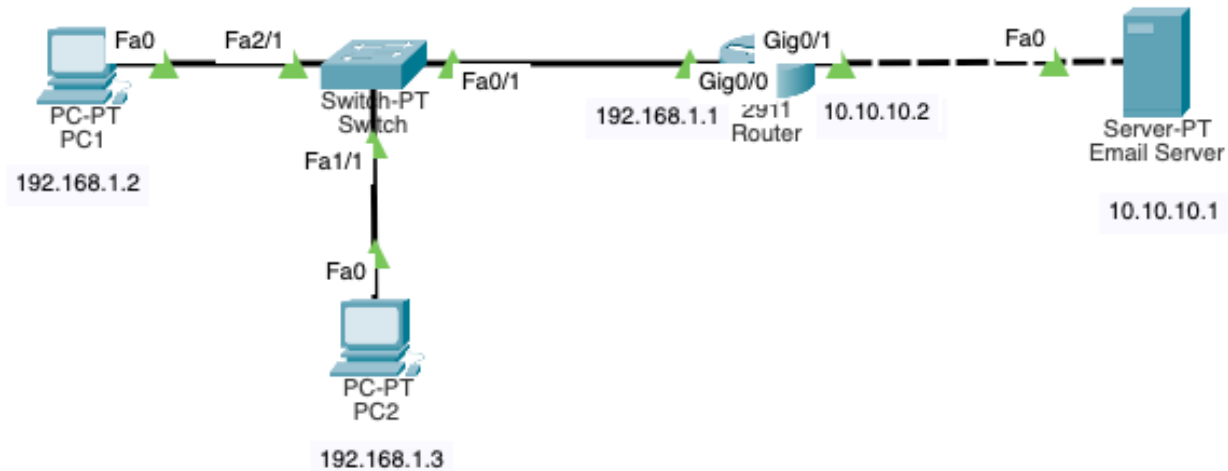
1. FastEthernet0 receives the frame.



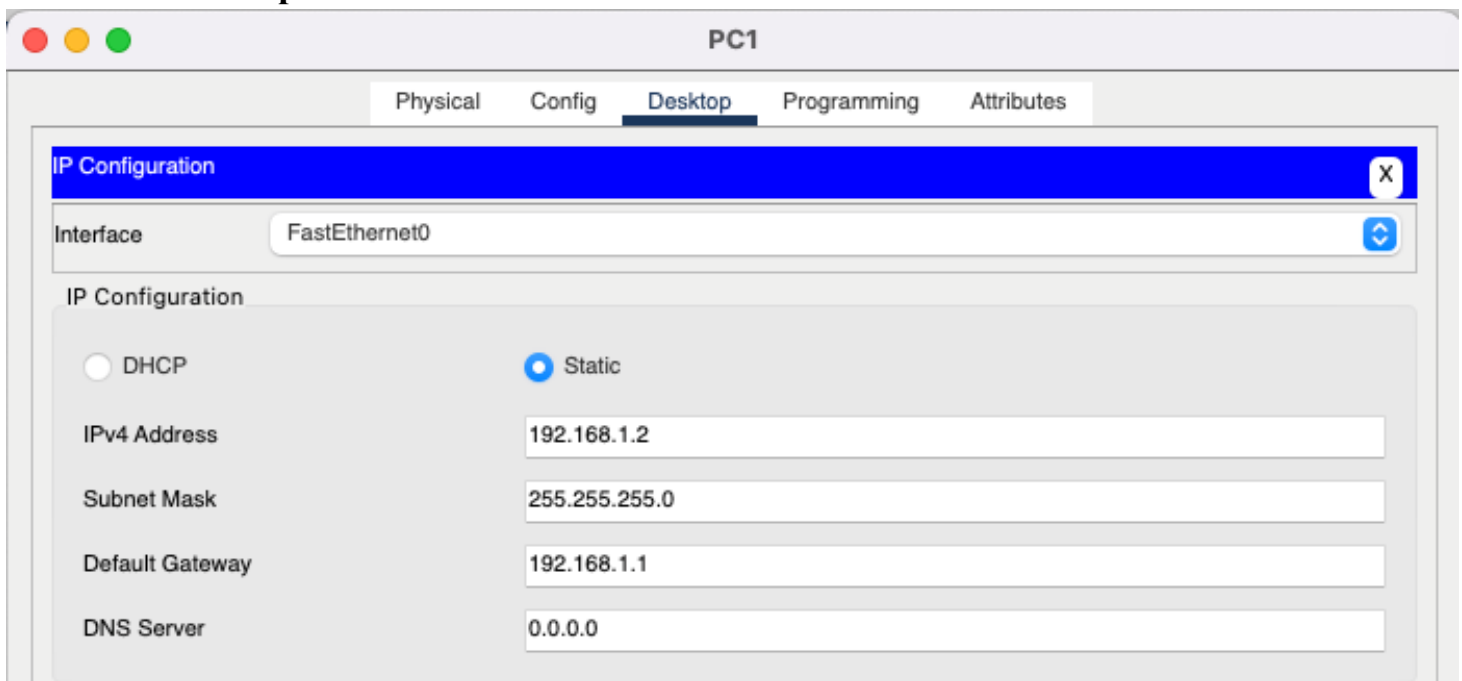
## Configuring an email server in Packet Tracer

### Step by Step Configuration:

#### 1. Network Topology



#### 2. Set up the IP Address



The image shows the **PC1** configuration window in Packet Tracer, specifically the **Desktop** tab. The **IP Configuration** window is open, showing the following settings:

- Interface:** FastEthernet0
- IP Configuration:**
  - ☐ DHCP
  - ☒ Static
- IPv4 Address:** 192.168.1.2
- Subnet Mask:** 255.255.255.0
- Default Gateway:** 192.168.1.1
- DNS Server:** 0.0.0.0

PC2

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☐ DHCP
☒ Static

IPv4 Address192.168.1.3

Subnet Mask255.255.255.0

Default Gateway192.168.1.1

DNS Server0.0.0.0

Email Server

PhysicalConfigServicesDesktopProgrammingAttributes

IP Configuration

IP Configuration

☐ DHCP
☒ Static

IPv4 Address10.10.10.1

Subnet Mask255.0.0.0

Default Gateway10.10.10.2

DNS Server0.0.0.0

### 3. Configure the Router

Router

Physical **Config** CLI Attributes

**GLOBAL**

Settings

Algorithm Settings

**ROUTING**

Static

RIP

**SWITCHING**

VLAN Database

**INTERFACE**

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

**GigabitEthernet0/0**

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000A.F32B.4A01

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Router

Physical **Config** CLI Attributes

**GLOBAL**

Settings

Algorithm Settings

**ROUTING**

Static

RIP

**SWITCHING**

VLAN Database

**INTERFACE**

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

**GigabitEthernet0/1**

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000A.F32B.4A02

IP Configuration

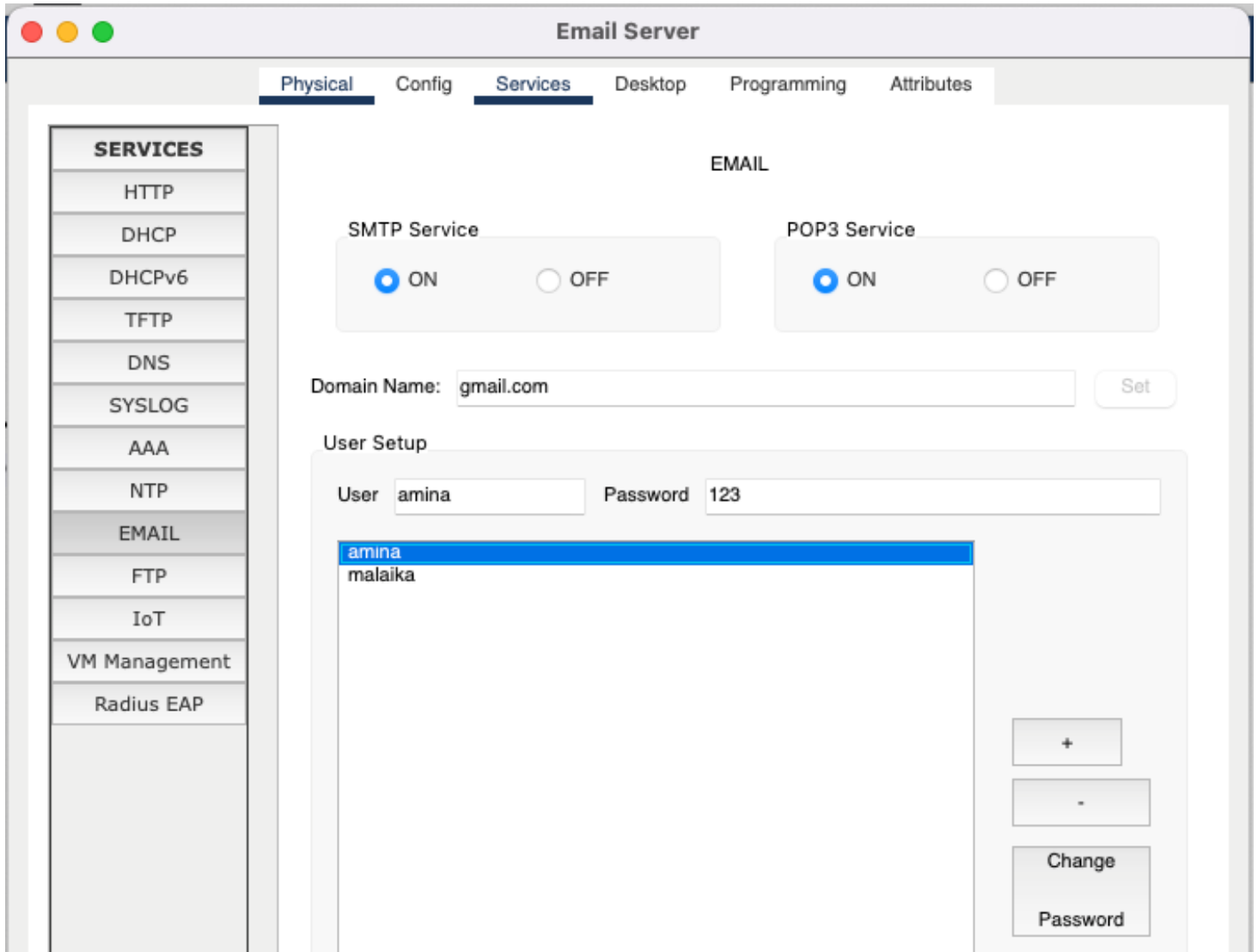
IPv4 Address 10.10.10.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

- This task can also be done with commands in the router terminal as well.
- Make sure that the correct ethernet port is chosen for both of the IP configuration.

#### 4. Configure the Mail Server



**Email Server**

Physical Config **Services** Desktop Programming Attributes

**SERVICES**

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**
- FTP
- IoT
- VM Management
- Radius EAP

**EMAIL**

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

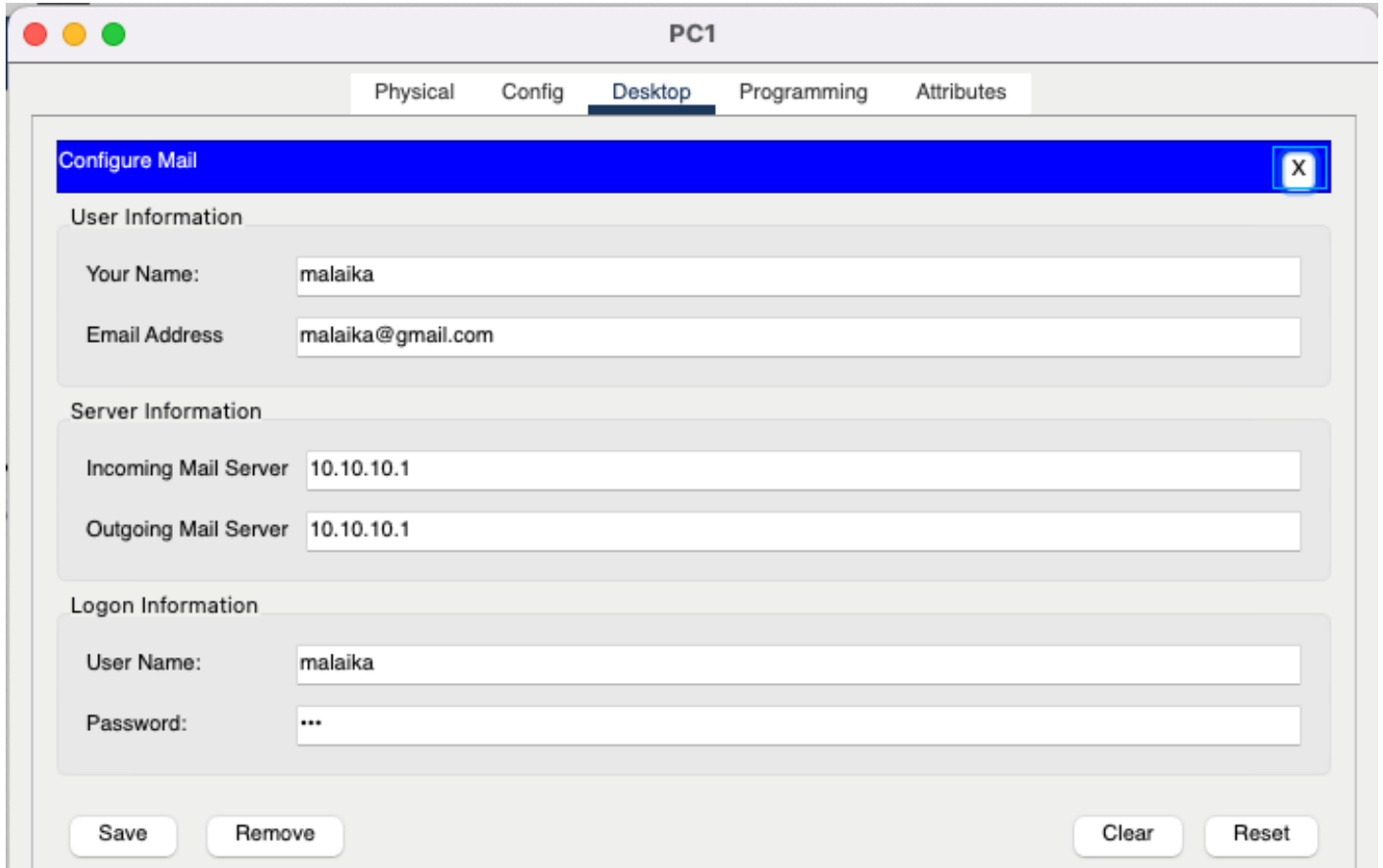
Domain Name:

User Setup

User  Password

- Navigate to the Services tab in the Email Server and go to 'EMAIL'.
- There first set the domain name (e.g. gmail.com)
- Then add users by providing User and Password and then hitting the plus '+' icon.
- This way you can register users in your Email Server to be used later.

## 5. Setting Mails on PC



PC1

Physical Config **Desktop** Programming Attributes

**Configure Mail** [X]

User Information

Your Name: malaika

Email Address: malaika@gmail.com

Server Information

Incoming Mail Server: 10.10.10.1

Outgoing Mail Server: 10.10.10.1

Logon Information

User Name: malaika

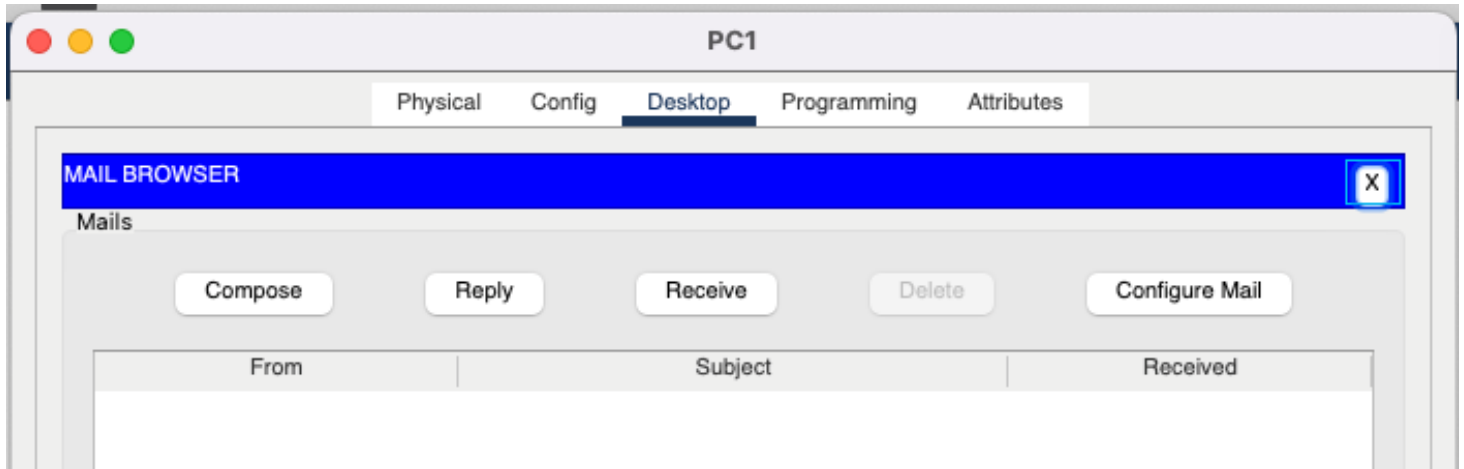
Password: \*\*\*

Save Remove Clear Reset

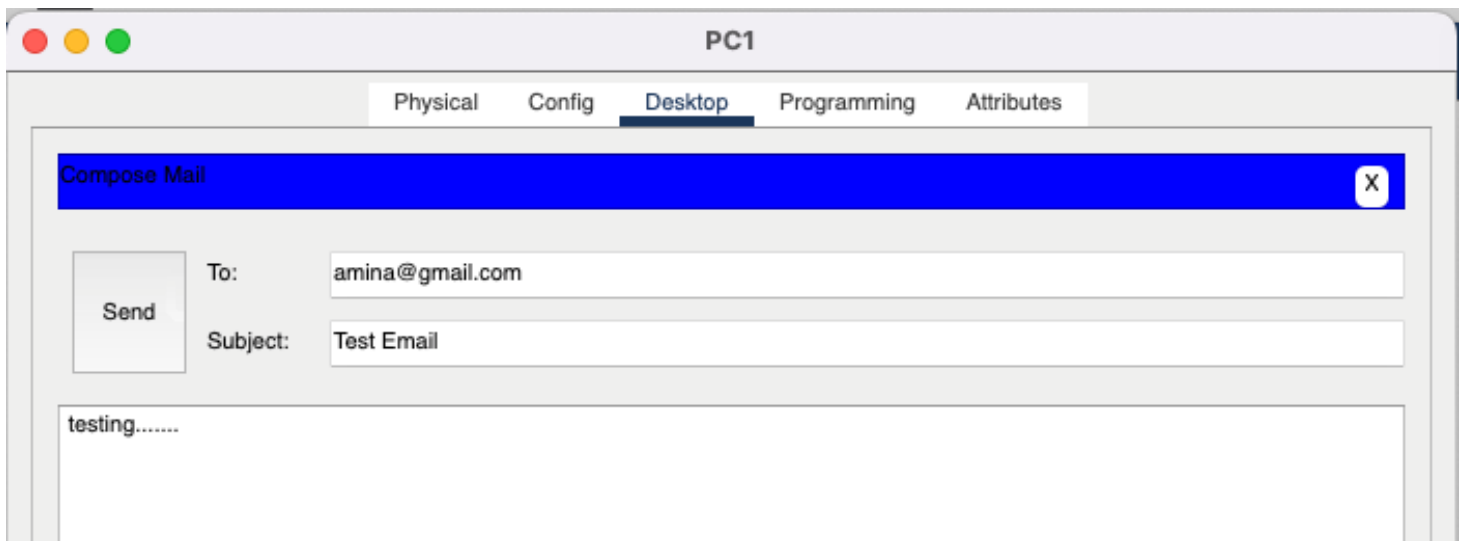
- Navigate to PC1 -> Desktop -> Email.
- Configure the email of the user you want to on this PC.
- Provide their name and email address. Keep in mind to use the specified domain name as before.
- The Logon information is to be the same as the one in the Email Server.
- Hit save and you're done.
- Do this for both PC1 and PC2.

## 6. Test Email Communication

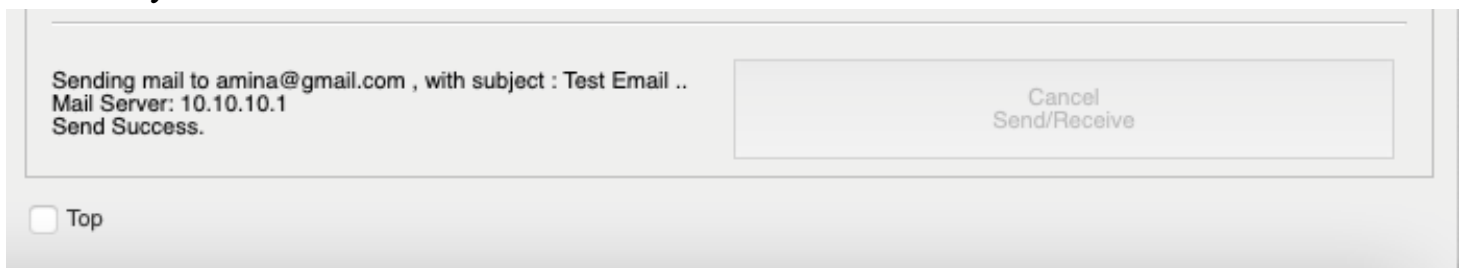
- To test if the email server has been configured correctly, click on the Compose button to write an email.



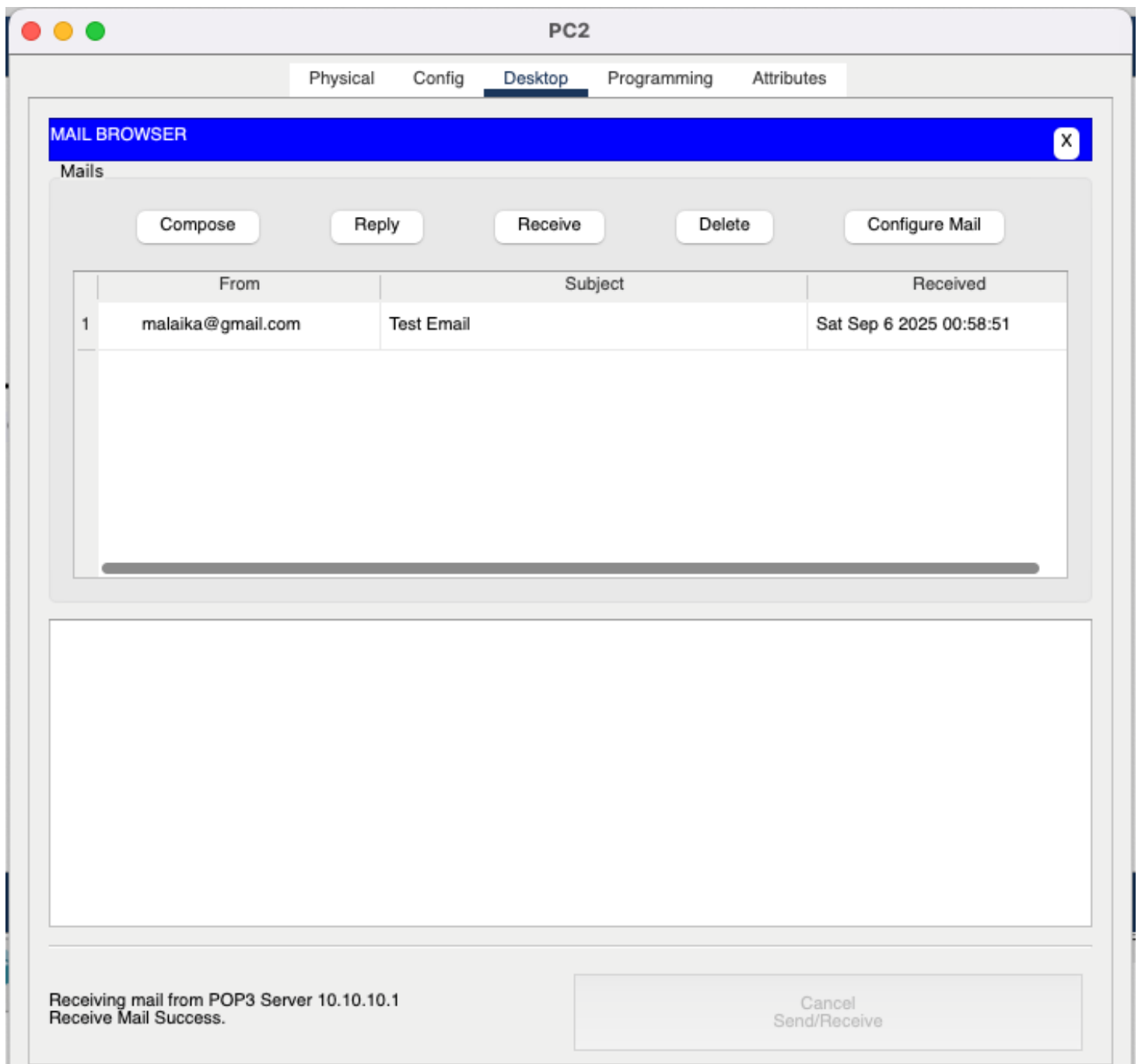
- Write the mail and hit send



- You will see whether or not the mail has been sent on the bottom left corner of your screen.



- Now open the second PC and go to the Email tab. Then click on receive to check if you got any mail or not.



- You will be shown all the mails you received. You can then reply back or delete them.

## Lab Task

### Task 1:

#### 1. Network Topology

- Set up 3 PCs and connect them to a single switch.
- A DNS Server and a Web Server, connecting them to another switch.
- Add a router to connect the two switches.
- Use appropriate cables to connect them all

#### 2. IPv4 Addressing

- Assign IP addresses to router interfaces and the networks it is connecting.

#### 3. DNS Server Configuration

- Turn on DNS Service.
- Add A records: name: [www.example.com](http://www.example.com) and Web Server IP address.
- Save changes.
- Go to each PC and DNS server IP.

#### 4. Web Server Configuration

- Turn on HTTP service.
- Edit index.html to add simple content (introduce yourself, Name, Roll number, hobbies and interests)

#### 5. Verification and Testing

- Test connectivity: ping from all PCs to Web Server and DNS Server.
- DNS Tests: From a PC open cmd and use **nslookup** [www.example.com](http://www.example.com) to verify resolution to the IP Address.
- Web Access: From a PC, open Web Browser and access [www.example.com](http://www.example.com) and verify the page loading.
- Simulation Mode: Switch to Simulation Mode, send a web request from a PC to [www.example.com](http://www.example.com). Capture PDUs and note layers (e.g., DNS query via UDP port 53, HTTP via TCP port 80).

### Task 2:

#### 1. Network Topology

- Design a Topology that includes:
  - 4 PCs in the CS Department, connected to a switch.
  - 4 PCs in the EE Department, connected to a switch.
  - One Router that connects both departments and a DNS server network.
  - A Mail Server in the CS and EE Department each.



- A central DNS Server connected through the router.
- Use appropriate cables to connect all devices.

## 2. IPv4 Addressing

- Divide the network into **three subnets**:
  - CS Department → 192.168.10.0/24
  - EE Department → 192.168.20.0/24
  - DNS Network → 192.168.30.0/24
- Assign IP addresses to:
  - Router interfaces (gateways for each subnet).
  - Mail Servers (CS & EE).
  - Central DNS Server.
  - PCs (each with correct gateway and DNS server).

## 3. DNS Server Configuration

- Configure the DNS Server with records for both domains:
  - mail.cs.edu → CS Mail Server
  - mail.ee.edu → EE Mail Server
- Ensure all PCs use this central DNS server.

## 4. Mail Server Configuration

- On the **CS Mail Server**, set the domain as mail.cs.edu and create at least two users.
- On the **EE Mail Server**, set the domain as mail.ee.edu and create at least two users.

## 5. Client Email Setup

- Configure the PCs to use the email accounts created on their respective mail servers.
- Ensure each PC is configured with the central DNS server.

## 6. Verification and Testing

1. **Connectivity Tests**
  - Verify that PCs can ping PCs in the other department.
  - Verify that PCs can ping the DNS server.
2. **DNS Tests**
  - From a CS PC, resolve mail.ee.edu.
  - From an EE PC, resolve mail.cs.edu.
3. **Email Tests**
  - Send an email from a CS user to an EE user.



- Reply back from an EE user to a CS user.

**Submission Guidelines:**

- Record each step by taking screenshots of it and attaching them to a pdf file.
- Submit a PDF document and Packet Tracer (.pkt) file.
- Naming convention of both the files: A\_i23xxx\_Lab4.